Fault maintenance ideas
故障维修思路

The possibility of blowing tin should be taken into account for the repaired motherboard
被人修过的主板，要把吹爆锡的可能性考虑进去

- Inspection and disassembly
  问排望测拆
  - 1: Ask the cause of the failure
    1：问引起故障的原因
  - 2 eliminate the possibility of failure caused by peripherals
    2:排除外设引起故障的可能性
  - 3: Observe the mainboard for obvious damage, deformation and repaired parts
    3：观察主板有无明显损坏变形及维修过的地方
  - 4: Measure the diode value and remove the obviously damaged components where there is obvious damage
    4：对于明显损坏的地方测量二极体值及拆除明显损坏的元件
  - 5: Focus on the inspection of places repaired by others
    5：对于别人维修过的地方重点检查
- Trigger fault
  触发故障
  - 1: Not triggered
    1:不触发
    - Measure the resistance and voltage of the starting line
      测开机线路阻值及电压
      - 1: Short circuit and remove relevant components
        1:短路拆相关元件
      - 2: No voltage check path
        2:无电压查通路
      - 3: Low voltage
        3:电压偏低
        - Remove relevant components
          拆相关元件
        - Replace the pull-up resistor and power supply
          更换上拉电阻及电源
        - give up
          放弃
  - 2 trigger constant current
    2:触发定电流
    - 1: Measure CPU reset voltage
      1：测量CPU复位电压
      - No voltage
        无电压
        - Measured resistance
          测阻值
        - Check the pull resistor and voltage.
          查上拉电阻及电压
      - Low voltage
        电压偏低
        - Measured resistance
          测阻值
        - Check the pull resistor and voltage.
          查上拉电阻及电压
        - Borrowed voltage, unable to pull up, maintenance completed
          外借电压，拉不起来维修结束
      - The voltage is normal
        电压正常
        - Connect the computer
          连电脑
    - 2: Connect the computer to see if it is connected
      2：连电脑看是否连机
      - Mandatory online
        强制性联机
        - No connection, no online repair
          不联，按不联机修
        - Online, press enter mode for repair
          联机，按进模式修
      - Progressive mode
        进模式
        - White DFU, version resistance or CPU bad
          白DFU，版本电阻或CPU坏
        - Replant hard disk
          重植硬盘
        - Brush the machine (repair according to the error reporting steps)
          刷机（根据刷机报错步骤进行维修）
  - 3: Trigger current 100 multi cycle restart
    3:触发电流100多循环重启
    - bus
      I2C0、I2C1总线
    - CPU
  - Trigger large current
    触发大电流
    - There is a short circuit in each power supply or conversion circuit of the power output
      电源输出的各路供电或者转换电路存在短路
    - Measure the power supply value of each circuit or touch it when starting up
      测量各路供电值或者开机用手摸

Note: when all faults are finally locked on the CPU, the CPU can be removed and installed on a good 1D board for test.
If the fault persists, the CPU is broken. If the fault is good, there is a problem at the bottom of the board

注：所有故障最后锁定在CPU上时，可把CPU拆下装在好的ID板上测试，如果故障依旧，说明CPU坏，如果故障好了，说明板底有问题

- Power failure
  接电故障
  - Electrical short circuit
    接电短路
    - 1: Battery supply circuit short circuit (front end)
      1:电池供电电路短路（前端）
    - 2: Main power supply short circuit (rear end)
      2:主供电短路（后端）
    - 3: Generation 7 starts with PP VDD_ Boost circuit (third end)
      3:7代开始有路PP_VDD_BOOST升压电路（第三端）

    Lock the fault line and burn the machine directly from the short circuit. If the power supply cannot burn out, you can burn the machine with battery.
    锁定故障出现的线路，直接从短路那一路烧机，如果电源烧不出可用烧机电池烧机。

    The voltage converter tube will be the first to get hot when it is short circuited. The voltage converter tube has no grounding pin, so it does not have a short circuit with the ground. It is only because the later stage short circuit can not bear so much current, so it will get hot
    接电短路的首先发烫的会是电压转换管，电压转换管是没有接地脚的，所以他不存在与地短路的情况，只是因为后级短路他承受不了那么大的电流，所以他会发热
  - Electric leakage (distinguish leakage or automatic triggering)
    接电漏电（区分漏电还是自动触发）
    - Automatic trigger constant current
      自动触发定电流
      - Measure whether the inductance around the CPU has voltage. If there is voltage, it will be triggered automatically
        测量CPU周边电感有无电压，有电压就是自动触发
      - Trigger constant current maintenance according to trigger fault
        触发定电流按触发故障维修
        - After the fault is repaired, the startup is triggered automatically. Check the relevant components of the startup line
          故障修复完成后还是自动触发开机的，查开机线相关元件
    - electric leakage
      漏电
      - 1: Those with leakage above 120mA can be touched by hand
        1:漏电120MA以上的可以用手摸
        - Some mainboards need to wait a minute or two before they feel hot
          有些主板接电需要等一两分钟才能感觉到发热
        - Find a hot place to smoke rosin
          找到发热的地方薰松香
      - 2: Leakage below 100mA is generally invisible.
        2:低于100MA以下的漏电一般是摸不出来的。
        - Buy infrared thermometer
          买红外测温仪
        - Trigger boot
          触发开机
          - Press not to trigger maintenance
            不开机按不触发维修
          - Can trigger
            能触发
            - Can enter the system: test various functions
              能进入系统：测各个功能
            - If the system cannot be accessed, press the trigger to repair the fault
              不能进入系统的按触发故障维修
- Boot failure
  开机故障
  - White apple restart
    白苹果重启
    - First eliminate the external configuration and confirm that the problem is on the motherboard
      先排除外配，确认问题出在主板上
    - Logic chip, 12C bus, system, hard disk, CPU
      逻辑码片、I2C总线、系统、硬盘、CPU
    - 6S and beyond, in addition to the above, there are baseband, tail plug, audio and CPU peripheral inductors
      6S及以后除上面外还有基带、尾插、音频、CPU周边电感
  - blurred screen
    花屏
    - CPU peripheral inductance
      CPU周边电感
    - GPU power supply and GPU SRAM power supply
      GPU供电与GPU_SRAM供电
    - CPU
  - Boot screen full screen English
    开机画面整屏英文
    - First eliminate the external configuration and confirm that the problem is on the motherboard
      先排除外配，确认问题出在主板上
    - compass
      指南针
    - Hard disk
      硬盘
    - CPU